

German Power Fair Value — Day-Ahead Forecasting Pipeline

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This pipeline implements a production-grade European power trading stack: SMARD ingestion (no auth), LLM-powered QA, two-stage probabilistic forecasting with conformal uncertainty, curve translation into tradable signals, a REST API, and Docker deployment. The 2024 German DA market is the test case — 457 negative-price hours (5.2%) and Nov/Dec Dunkelflaute power prices reaching €820/MWh and €900/MWh — the highest in 18 years. This pipeline forecasts DA prices across both regimes and translates outputs into tradable signals.

1. Data Ingestion and Quality

Source: SMARD (smard.de) — Bundesnetzagentur electricity market platform. Data retrieved from ENTSO-E under EU Regulation 543/2013. Equivalent data quality, no API key required, fully public.

Programmatic API endpoints:

Series	Filter ID	Endpoint Pattern	Unit
DA Price DE-LU	4169	/chart_data/4169/DE-LU/	EUR/MWh
Total Load	410	/chart_data/410/DE/	MW
Wind Onshore	123	/chart_data/123/DE/	MW
Wind Offshore	3791	/chart_data/3791/DE/	MW
Solar PV	125	/chart_data/125/DE/	MW

Base URL: `https://www.smard.de/app/chart_data/{filter}/{region}/{filter}_{region}_hour_{ts}.json`
Response: `[[unix_ms, value], ...]` arrays converted to UTC-indexed hourly Series.

Dataset: 26,304 hours, 2022-01-01 to 2024-12-31.
Timezone: UTC internal storage; CET/CEST for display. Spring-forward gap interpolated; autumn duplicate averaged. Assert: exactly 8,784 rows for 2024 — passes.

Structural breaks: `post_ukraine_war` (2022-02-24); `post_nuclear_phaseout` (2023-04-15, DE nuclear = 0 MW).

LLM Data QA (Gemini 2.0 Flash): 20 electricity-market-specific validation rules proposed programmatically and executed as Python boolean conditions (solar zero 20:00–05:00 CET, load 20–100 GW, price – 500 to 3,000 EUR/MWh, nuclear = 0 after phase-out, Dunkelflaute logic). QA score: 97.1/100 — PASS. This reduces manual rule authoring from hours to seconds while producing domain-specific rules a generalist engineer would miss.

2024 verified facts: Mean DA price €78.51/MWh; negative hours 457 (5.2%); Dunkelflaute

detected: Nov 2–7 2024 (111 Dunkelflaute hours within 144h window), Dec 12–14 2024 (45 Dunkelflaute hours within 72h window).

2. Forecasting and Model Validation

Target: Hourly DA prices for delivery day + 1. Delivery-period aggregates derived from the hourly distribution.

Fundamental insight: German prices are driven by residual load = load – wind – solar. High residual load → gas marginal → prices spike. Low or negative → renewables marginal → prices collapse. Features capture this at multiple timescales.

67 features across 8 groups: fundamental supply/demand (11, led by residual_load), fuel/carbon (7, including clean spark spread), calendar with cyclic encoding (16), lag features shifted minimum 24h (8, price_lag_168h is strongest predictor at $r=0.71$), rolling statistics (10), interaction terms (6), regime features (8, including dunkelflaute_severity), and structural break dummies (3). Leakage assertion: no feature shows correlation above 0.95 with future target.

Results:

Model	2024 MAE	Skill vs Naive
Seasonal Naive	~32 EUR/MWh	baseline
Ridge Regression	37.91 EUR/MWh	-15.5%
XGBoost + Two-Stage	25.37 EUR/MWh (RMSE: 40.0)	+22.7%

MAE by year (walk-forward 2022–2024):

Year	MAE	Context
2022	80.18	Ukraine crisis — prices €200–400/MWh
2023	52.54	Transition year
2024	25.37	Primary benchmark year

Negative-price recall: **87.7%** of 457 actual negative-price hours correctly flagged as high-risk — directly relevant to renewable cannibalization trading. Directional accuracy: **82.9%**. Two-stage architecture: binary classifier routes 17.2% of hours to an extreme-event regressor trained with price-proportional sample weights; a dedicated negative-price classifier (scale_pos_weight = 18.2, summer recall 91.1%, full-year 88.0%) feeds probability as a feature into the main model.

Validation: 52-window expanding walk-forward; min train 365 days; step 7 days; test full 2024 (8,784 hours); zero shuffling, zero future leakage.

Conformal Prediction (O'Connor et al., 2025): Regime-conditioned split conformal prediction with distribution-free coverage — no Gaussian assumptions. Empirical 90% coverage: **91.0%**. Full 2024 out-of-sample predictions in submission.csv (8,784 rows): point forecast, 7 quantiles, conformal intervals, regime label, and Dunkelflaute risk score per hour.

3. Prompt Curve Translation

Horizon	Definition	Instrument
Prompt Day Baseload	Mean 24h	DE DA base block
Prompt Day Peak	Mean hours 08–20 Mon–Fri	DE DA peak block
Prompt Week Base	7-day mean	EEX Week Base Future
Prompt Month Base	Calendar month mean	EEX Month Base Future

All with 80% and 95% conformal prediction intervals.

Signal: LONG if fair-value estimate (FV) $>$ reference $+ 0.5 \times \text{CP80_width}$; SHORT if FV $<$ reference $- 0.5 \times \text{CP80_width}$; NEUTRAL otherwise. HIGH conviction if CP80 width $<$ 80% of 7-day mean.

Regime 3 (Dunkelflaute, residual load above 55GW and wind below 5GW): LONG prompt peak. Nov 2024 event drove DA prices to €145/MWh — gas peakers earned a disproportionate share of annual wholesale revenue over 111 hours. Regime 0 (renewable glut, penetration above 80% on summer weekends): SHORT prompt baseload, long battery charge/discharge spread.

Invalidation: wind revision $>$ 20%; TTF move $>$ €3/MWh overnight; French nuclear drop $>$ 2,000 MW; DA clearing $>$ 2σ from forecast; residual load change $>$ 15%; unplanned outage $>$ 1,000 MW; regime flip pre-delivery; conformal band expands $>$ 50% vs 7-day mean.

Simplified 2024 backtest: 326 trades, 82.2% win rate, +10,272 EUR/MW P&L (illustrative — no transaction costs).

4. AI-Accelerated Workflow

Three Gemini 2.0 Flash integrations — all programmatic, logged to JSONL, auditable. Also exposed via FastAPI REST API (6 endpoints, Swagger at /docs) and Docker.

Component 1 — LLM Data QA: Gemini receives schema plus 10 sample rows plus statistics and proposes 20 electricity-market-specific validation rules executed programmatically as Python boolean conditions. Component 2 — Market Commentary: 150–180 word Bloomberg-style note generated from 24 pipeline metrics only, with anti-hallucination guard cross-checking every number at ± 0.5 tolerance. Component 3 — Ingestion Config: field documentation converted to structured JSON ingestion config, eliminating manual field mapping hardcoding. All prompts, responses, and outputs logged to JSONL under outputs/logs/.

Component	Logged	Hallucination Check	On Failure
QA Rules	Yes — JSONL	N/A	Log + continue
Commentary	Yes — JSONL	Yes — ± 0.5 tolerance	Flag + log
Config Gen	Yes — JSONL	N/A	Fallback defaults

References: O'Connor et al. (2025); Marcjasz et al. (2023); Weron (2014); Wood Mackenzie (2025); Timera Energy (2025); FfE (2026).